Air pollution impacts on disadvantaged communities in St. Louis, Missouri

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Topic of Choice:

I attended Washington University in St. Louis, in St. Louis Missouri for my undergrad eduation. This school has a very wealthy student body located in a city that struggles with a history of redlining, pollution, and other socioeconomic and environmental issues. I want to learn more about the issues facing the disadvantaged populations of St. Louis, so that is what I focused on for this assignment; specifically the unqeual distribution of air pollution from the city's industrial roots.

The American Lung Association's 2026 "State of the Air" report (https://www.lung.org/research/sota/city-rankings/msas/st-louis-st-charles-farmington-mo-il) ranked the St. Louis metro area as one of the worst in the country for ozone pollution as well as particle pollution.

I wanted to look at whether this pollution disproportionately impacted disadvantaged populations (namely people of color, limited english speaking households, low income, unemployed, or those with less than a high school education).

Reading in the data, and selecting subset of interest

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
Warning: package 'tidyr' was built under R version 4.4.3
```

```
Warning: package 'purrr' was built under R version 4.4.3
```

```
Warning: package 'dplyr' was built under R version 4.4.3

Warning: package 'lubridate' was built under R version 4.4.3

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.2     v tibble    3.3.0
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(sf)
```

```
Warning: package 'sf' was built under R version 4.4.3

Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(here)
```

```
here() starts at /Users/oMac/Desktop/meds_stuff/eds223 repos folder/eds223-
homeworks/eds-223-homework-1
```

```
library(tmap)

# Read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data", "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/oMac/Desktop/meds_stuff/eds223 repos folder/eds223-homeworks/eds-
223-homework-1/data/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```r
# Filter to Missouri and Illinois, since the greater St. Louis area crosses state borders
mo_and_so_il <- ejscreen %>%
  dplyr::filter(ST_ABBREV %in% c("MO", "IL"))

# Filter to counties making up the greater stl area


saint_louis <- mo_and_so_il %>%
  dplyr::filter(
    (CNTY_NAME %in% c("St. Louis City") & ST_ABBREV %in% c("MO")) |
    (CNTY_NAME %in% c("St. Clair County") & ST_ABBREV %in% c("IL")) |
    (CNTY_NAME %in% c("Madison County") & ST_ABBREV %in% c("IL")) |
    (CNTY_NAME %in% c("St. Louis County") & ST_ABBREV %in% c("MO"))
  )
```

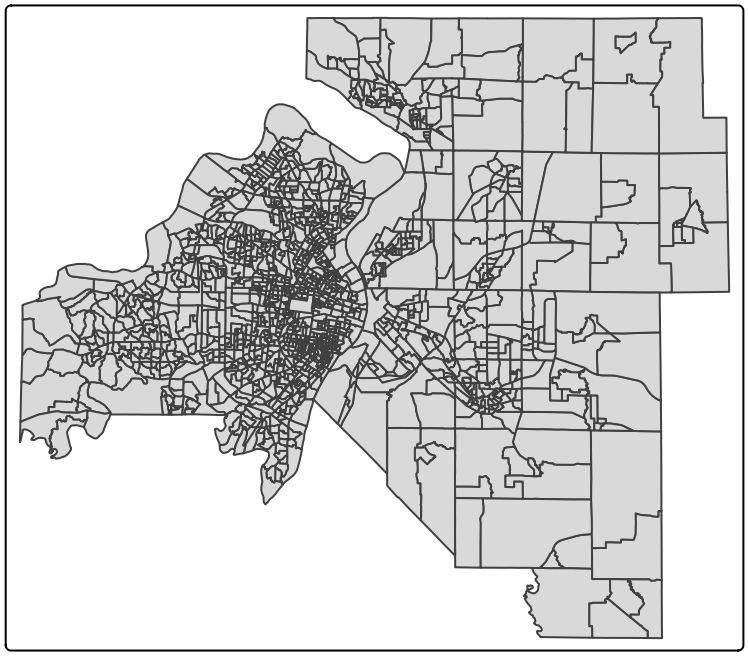
Make base map of area of interest.

The greater St. Louis area includes St. Louis City and St. Louis County in Missouri, as well as St. Clair and Madison counties in Illinois:

```r
# Make base map

stl_shape <- st_as_sf(saint_louis)

base_map <- tm_shape(stl_shape) +
    tm_polygons()
base_map
```



Looking specifically at percentile measures (>90th percentile, to aid comparison across various measures), how do these maps of air pollution compare to maps of socioeconomic disadvantage factors?

```
# Filter for areas where the majority of the population falls into at least one socioeconomi

dis_area <- saint_louis |>
    dplyr::filter(PEOPCOLORPCT > 0.5 | LOWINCPCT > 0.50 | UNEMPPCT > 0.5 | LINGISOPCT > 0.5

dis_shape <- st_as_sf(dis_area)
```

Compare disadvantaged community map to maps of air pollution factors

Look at areas that fall into the above categories, and also are above the 90th percentile for any of the air pollution factors in the dataset (PM2.5 exposure, ozone exposure, diesel particulate matter exposure, cancer risk due to airborne particulate matter, respiratory illness risk due to airborne particulate matter, and toxic releases to air).

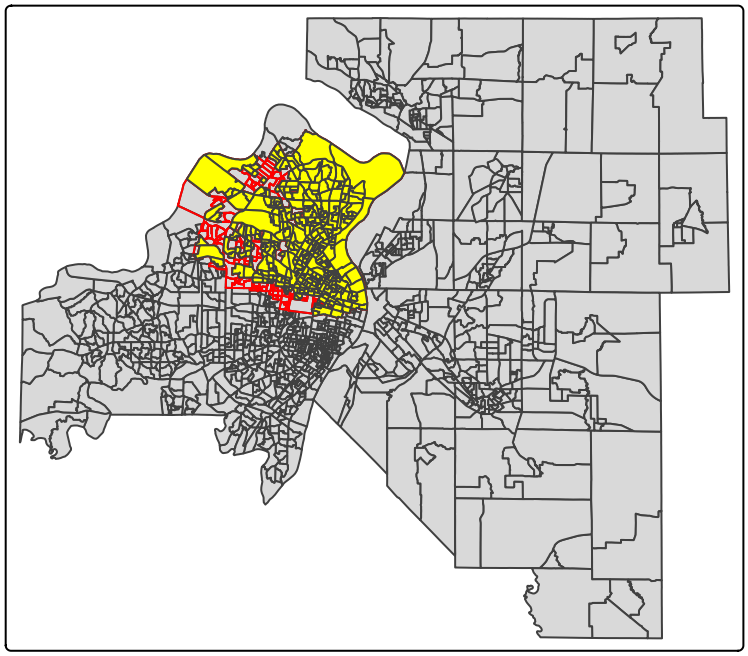
```
# Look at areas that fall under the above categories AND are in the 90th percentile for PM2.5
dis_pm25 <- dis_area |>
  dplyr::filter(P_PM25 > 90)
all_pm25 <- saint_louis |>
```

```
  dplyr::filter(P_PM25 > 90)

# Map it!
dis_pm25_shape <- st_as_sf(dis_pm25)
all_pm25_shape <- st_as_sf(all_pm25)

pm25_map <- tm_shape(stl_shape) +
    tm_polygons() +
  tm_shape(all_pm25_shape) +
    tm_lines(col="red") +
  tm_shape(dis_pm25_shape) +
    tm_polygons(fill = "yellow")

pm25_map
```



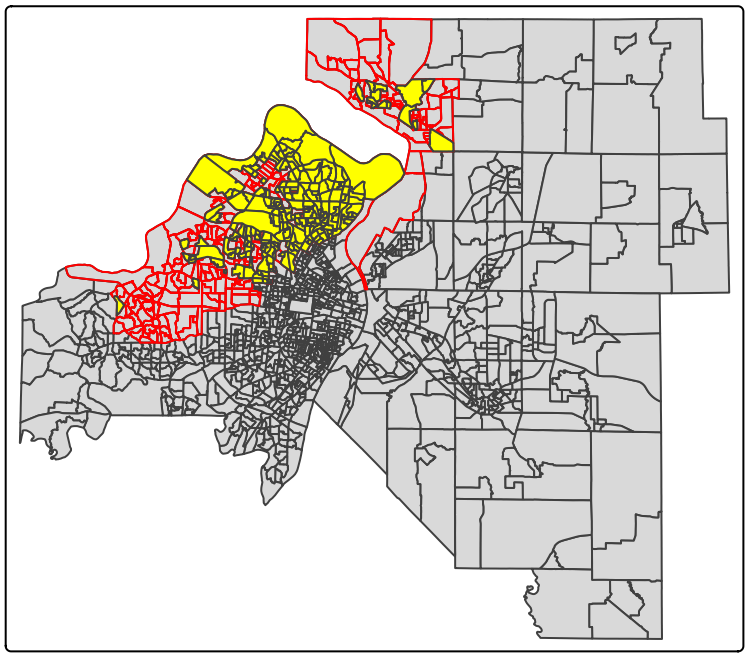
```
# Look at areas that fall under the above categories AND are in the 90th percentile for ozone
dis_ozone <- dis_area |>
  dplyr::filter(P_OZONE > 90)
all_ozone <- saint_louis |>
  dplyr::filter(P_OZONE > 90)

# Map it
```

```
dis_ozone_shape <- st_as_sf(dis_ozone)
all_ozone_shape <- st_as_sf(all_ozone)

ozone_map <- tm_shape(stl_shape) +
    tm_polygons() +
  tm_shape(all_ozone_shape) +
    tm_lines(col="red") +
  tm_shape(dis_ozone_shape) +
    tm_polygons(fill = "yellow")

ozone_map
```

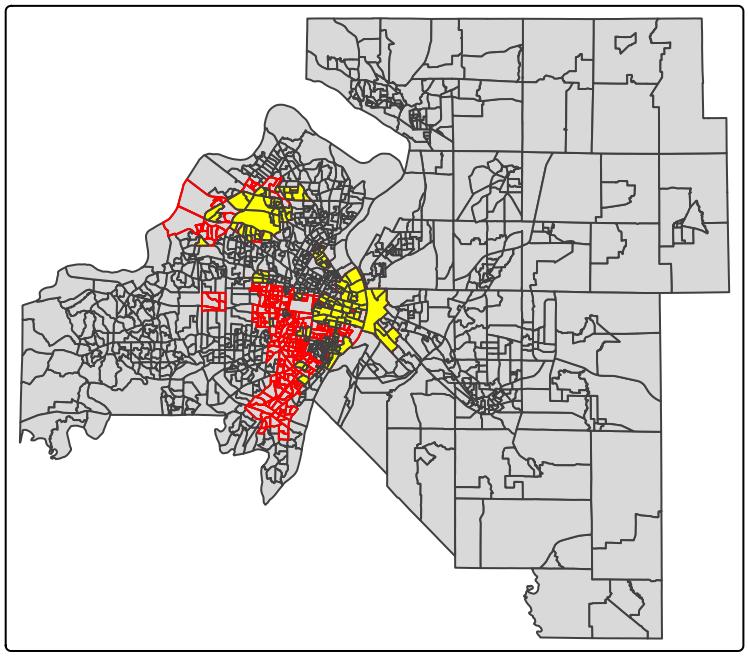


```
# Look at areas that fall under the above categories AND are in the 90th percentile for diese
dis_diesel <- dis_area |>
  dplyr::filter(P_DSLPM  > 90)
all_diesel <- saint_louis |>
  dplyr::filter(P_DSLPM  > 90)

# Map it
dis_diesel_shape <- st_as_sf(dis_diesel)
all_diesel_shape <- st_as_sf(all_diesel)
```

```
diesel_map <- tm_shape(stl_shape) +
    tm_polygons() +
  tm_shape(all_diesel_shape) +
    tm_lines(col="red") +
  tm_shape(dis_diesel_shape) +
    tm_polygons(fill = "yellow")

diesel_map
```



```
# Look at areas that fall under the above categories AND are in the 90th percentile for cance
dis_cancer <- dis_area |>
  dplyr::filter(P_CANCER  > 90)
all_cancer <- saint_louis |>
  dplyr::filter(P_CANCER  > 90)

# Map it
dis_cancer_shape <- st_as_sf(dis_cancer)
all_cancer_shape <- st_as_sf(all_cancer)

cancer_map <- tm_shape(stl_shape) +
    tm_polygons() +
  tm_shape(all_cancer_shape) +
```

```
    tm_lines(col="red") +
  tm_shape(dis_cancer_shape) +
    tm_polygons(fill = "yellow")

cancer_map
```

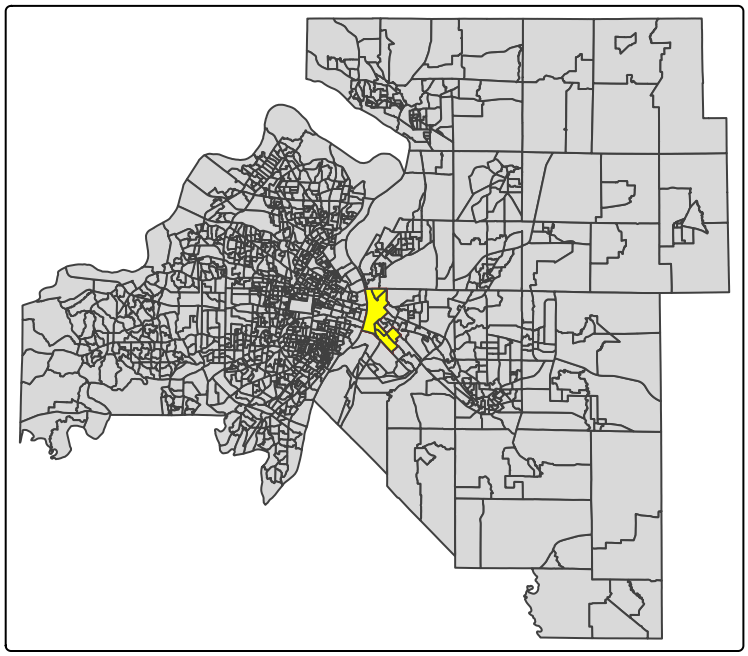


```
# Look at areas that fall under the above categories AND are in the 90th percentile for respi
dis_resp <- dis_area |>
  dplyr::filter(P_RESP  > 90)
all_resp <- saint_louis |>
  dplyr::filter(P_RESP  > 90)

# Map it
dis_resp_shape <- st_as_sf(dis_resp)
all_resp_shape <- st_as_sf(all_resp)

resp_map <- tm_shape(stl_shape) +
    tm_polygons() +
  tm_shape(all_resp_shape) +
    tm_lines(col="red") +
  tm_shape(dis_resp_shape) +
    tm_polygons(fill = "yellow")
```

```
resp_map
```

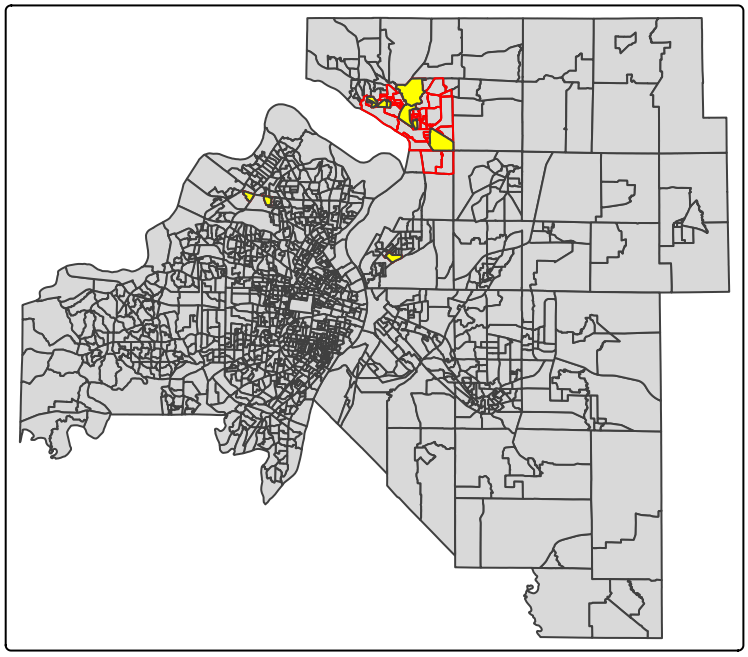


```
# Look at areas that fall under the above categories AND are in the 90th percentile for Toxi
dis_rsei <- dis_area |>
  dplyr::filter(P_RSEI_AIR  > 90)
all_rsei <- saint_louis |>
  dplyr::filter(P_RSEI_AIR  > 90)

# Map it
dis_rsei_shape <- st_as_sf(dis_rsei)
all_rsei_shape <- st_as_sf(all_rsei)

rsei_map <- tm_shape(stl_shape) +
    tm_polygons() +
  tm_shape(all_rsei_shape) +
    tm_lines(col="red") +
  tm_shape(dis_rsei_shape) +
    tm_polygons(fill = "yellow")

rsei_map
```



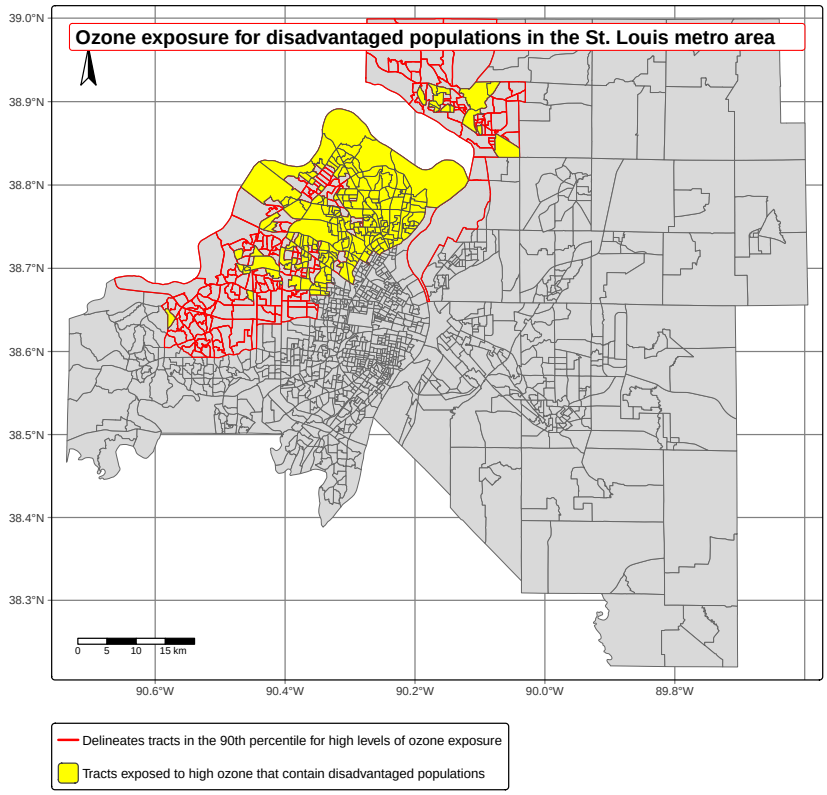
Make these maps look pretty!

```
# The ozone map shows the strongest relationship between air pollution exposure and disadvant
ozone_map <- tm_shape(stl_shape) +
    tm_graticules() +
    tm_polygons(fill = "grey85", col = "grey40", lwd = 0.3) +
  tm_shape(all_ozone_shape) +
    tm_lines(col="red") +
  tm_shape(dis_ozone_shape) +
    tm_polygons(fill = "yellow", col = "grey30", lwd = 0.5) +
  tm_add_legend(
    type = "lines",
    labels = "Delineates tracts in the 90th percentile for high levels of ozone exposure",
    col = "red",
    lwd = 2,
    title = ""
  ) +
  tm_add_legend(
    type = "polygons",
    labels = "Tracts exposed to high ozone that contain disadvantaged populations",
    fill = "yellow",
```

```
    title = ""
  ) +
  tm_compass(position = c("top", "left")) +
  tm_scalebar(position = c("bottom", "left")) +
  tm_title(text = "Ozone exposure for disadvantaged populations in the St. Louis metro area"

ozone_map
```

```
[plot mode] fit legend/component: Some legend items or map compoments do not
fit well, and are therefore rescaled.
i Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```

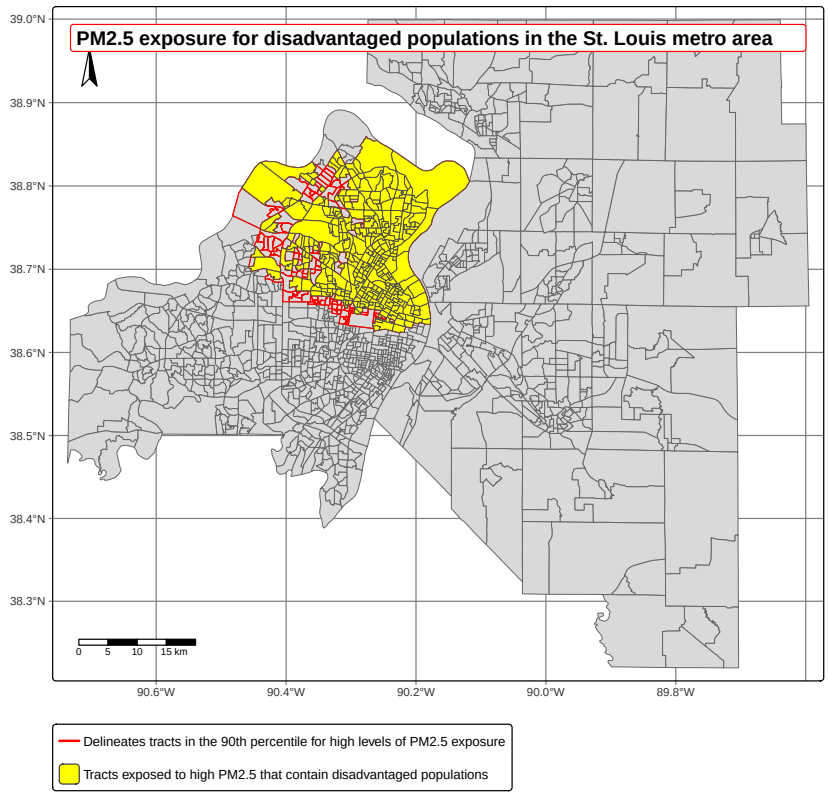



```
# The PM25 map also shows a strong relationship, so we can expand on this one as well

pm25_map <- tm_shape(stl_shape) +
    tm_graticules() +
    tm_polygons(fill = "grey85", col = "grey40", lwd = 0.3) +
```

```
  tm_shape(all_pm25_shape) +
    tm_lines(col="red") +
  tm_shape(dis_pm25_shape) +
    tm_polygons(fill = "yellow", col = "grey30", lwd = 0.5) +
  tm_add_legend(
    type = "lines",
    labels = "Delineates tracts in the 90th percentile for high levels of PM2.5 exposure",
    col = "red",
    lwd =2,
    title = ""
  ) +
  tm_add_legend(
    type = "polygons",
    labels = "Tracts exposed to high PM2.5 that contain disadvantaged populations",
    fill = "yellow",
    title = ""
  ) +
  tm_compass(position = c("top", "left")) +
  tm_scalebar(position = c("bottom", "left")) +
  tm_title(text = "PM2.5 exposure for disadvantaged populations in the St. Louis metro area"

pm25_map
```



Conclusions

As shown by these two maps, air pollution in St. Louis disproportionately affects disadvantaged populations (defined here as people of color, limited english speaking households, low income, unemployed, or those with less than a high school education).

The major contributor here is PM2.5, which is defined as small inhalable airborne particles with a diameter smaller than 2.5 micrometers (EPA, https://www.epa.gov/pm-pollution/particulate-matter-pm-basics). As shown by figure 1 (the PM2.5 map), areas with high annual PM2.5 levels ("high" here is defined as above the 90th percentile nationwide) almost completely overlap with areas matching at least one of the previously stated disadvantaged populations categories. This shows that PM2.5 air pollution disproportionately effects disadvantaged populations in the St. Louis metro area.

Ozone levels in St. Louis are also an issue. Ozone (O3) is an airborne pollutant with significant health impacts (EPA, https://www.epa.gov/ground-level-ozone-pollution/ground-level-ozone-basics). As shown in figure 2 (the ozone map), areas with high ozone levels ("high" here is defined as above the 90th percentile nationwide) overlap significantly with areas matching at least one of the previously stated disadvantaged populations categories. While not as

strong of a relationship as the PM2.5, it is clear here that disadvantaged communities are still disproportionately affected by ozone air pollution in the St. Louis metro area.

Data Citations

Data source: The data used for this project was originally sourced from the EPA's EJScreen: Environmental Justice Screening and Mapping tool. This site has since been taken down, but much of the data can still be accessed at EJScreen (https://pedp-ejscreen.azurewebsites.net).